

Jennifer Luo

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EDUCATION

Tufts University

Bachelor of Science in Computer Science
GPA: 3.73/4.0

Medford, MA

Aug 2023 - Present

EXPERIENCE

JumboCode

Oct. 2023 – Present

Head of Project Management, Full-Stack Developer

Medford, MA

- **Led two 14-member development teams** to deliver a full-stack library web app for non-profits, applying client-server architectures and leveraging REST APIs for data exchange (ReactJS, NextJS, TypeScript, Prisma, PostgreSQL).
- **Engineered and implemented RBAC** and audit logging mechanisms to safeguard tutor and volunteer data, ensuring multi-role access in compliance with privacy requirements.
- **Managed 12 project teams** as Head of Project Management, coaching PMs on agile system design and leadership, facilitating efficient project scope definition, execution, and timely client deliverables.
- **Developed backend infrastructure** for nonprofit *HomeStart* by designing scalable database schemas and RESTful API endpoints with optimized performance and data integrity (MongoDB, Prisma, Node.js).

Tufts University

Jan. 2024 – Present

CS Teaching Assistant

Medford, MA

- **Ran weekly labs** of 20+ students and held office hours to reinforcing algorithmic concepts such as dynamic programming and graphs, improving average exam scores by **4%**.
- **Evaluated 40+ student algorithm implementations weekly** for correctness across all input cases and adherence to theoretical complexity bounds, providing detailed technical feedback.

Nemitz Lab, Tufts University

Jan 2026 – Present

Software Researcher

Somerville, MA

- Developed a full-stack **self-driving lab** interface in to automate soft robotic actuator fabrication and characterization
- Designed **PostgreSQL** schema for experiment tracking and built a **computer vision** pipeline using OpenCV and an RGB-D camera for shadow-invariant actuator segmentation and curvature regression

PROJECTS

Projects

Full-Stack, ML/LLM, Machine Structure

- **HackMIT "AI Assistant" – Smart-glasses aid for dementia.** Combined YOLOv3 CV + LLM reasoning + speech to deliver context cues; knowledge-graph feedback loop improved prompt accuracy over 25+ test cases.
- **White.** Full-stack text editor web app with Clerk authentication and Notion-style block editing (React, Next.js, Node.js, Prisma, PostgreSQL on AWS RDS, TailwindCSS, Figma) [Github]
- **Movie Recommendation System.** Created a movie recommendation system using SGD and a latent factor/collaborative filtering model for recommendation that achieved RMSE of **0.934** on the validation set. [Doc]
- **File Compression.** Built an image file compressor and decompressor in C by transforming color spaces and performing lossy compression, reducing file size by **50%** while maintaining visual fidelity.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, Typescript, JavaScript, Java, MATLAB, HTML/CSS

Frameworks: ReactJS, NextJS, TailwindCSS, NodeJS(Express), ThreeJS, scikit-learn, Prisma, YOLOv3

Databases: PostgreSQL, Firebase, MongoDB, Supabase, NeonTech, Mentra Cloud DB

Cloud & Deploy: AWS, Vercel, Cloud Operations

Tools: Git, Linux, Linear, Clerk, Wireshark, DevOps Tools

Core Competencies: Software Development, APIs, Agile Methodology, Project Management Tools